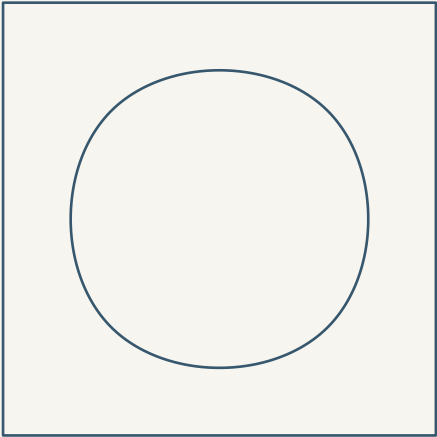


# From host surface to controlled growth

One prepared segment network. Spatially assigned settings. Four coupled goals. A separate check-and-save workflow.

01 / PREPARE THE GEOMETRY

Define where growth can occur.



The host supplies a seed boundary, a constraint mesh, and field bounds.

SOURCE EXPORT / HOST + INITIAL SEED

SEED PATH

Build → rebuild → pull to host

Generate a host-derived closed seed; rebuild it, then pull the rebuilt curve back onto the face.



SEGMENT PATH

Divide → shift → connect

Form a closed, fixed segment network. The saved definition starts with 1,500 segments.

G

PARALLEL HOST PATH / MESH BREP

Host mesh → OnMesh goal + field domain. The physics mesh is separate from preview refinement.

H

02 / ASSIGN LOCAL BEHAVIOUR

Translate design intent into settings.

DESIGN CONTROLS

Uniform · gradient · point · curve

Blend and contrast set the field effect. Axis / bias set the gradient; reach and background shape point and guide-curve influence.

H



DENSITY FIELD / EMBEDDED PYTHON

Sample the initial segments

Evaluate the field at segment midpoints. Smooth around the closed network, then map each value to a local control multiplier.

P



THREE PER-SEGMENT SETTINGS

|                  |                       |
|------------------|-----------------------|
| Rest length      | Expansion target      |
| Collision radius | Local separation      |
| Bending strength | Resistance to folding |

ASSIGNMENT LIFETIME

Values stay attached to their initial segments. Moving particles do not resample the spatial field.

03 / RELAX THE NETWORK

Let competing goals develop the form.

KANGAROO / FOUR PARALLEL GOALS

G

A Length(Line)  
Seek assigned rest lengths.

B Collider  
Separate nearby segments.

C Angle  
Resist adjacent-line bending.

D OnMesh  
Constrain particles to the host mesh.

All goals act on the same particle system.

BOUNCYSOLVER  
Update positions + momentum

Repeat internally while running

04 / INSPECT AND CAPTURE

Read the result and preserve evidence.



Developed growth  
Actual supplied vector geometry. Not a new run.

SOURCE EXPORT / DEVELOPED STATE

SOLVER VERTICES → NURBS CURVE

Construct the smooth output

The closed display curve is reconstructed from solver vertices after particle-based constraints.



PAUSE RUN → CHECK

Assess the resulting curve

Validity, closure, length and sampled diagnostics.



SAVE RUN / OPTIONAL

Keep geometry + evidence

geometry.3dm / run.json / curve\_samples.csv  
Archive the matching Grasshopper file separately.

READ THE BRANCHES

H + G / SHARED GEOMETRY

A single geometric reference

H: host mesh → OnMesh goal and field domain.  
G: initial segments / adjacent pairs → goal geometry;  
segment endpoints → OnMesh particle positions.

P / FULL-FORM FIELD PREVIEW

Same field function, different sampling

Evaluate the raw field on a refined copy of the host mesh.  
This happens before cyclic segment smoothing.  
The preview is a reference; it never drives the solver.

E / INTERPRETING THE EVIDENCE

What the workflow establishes

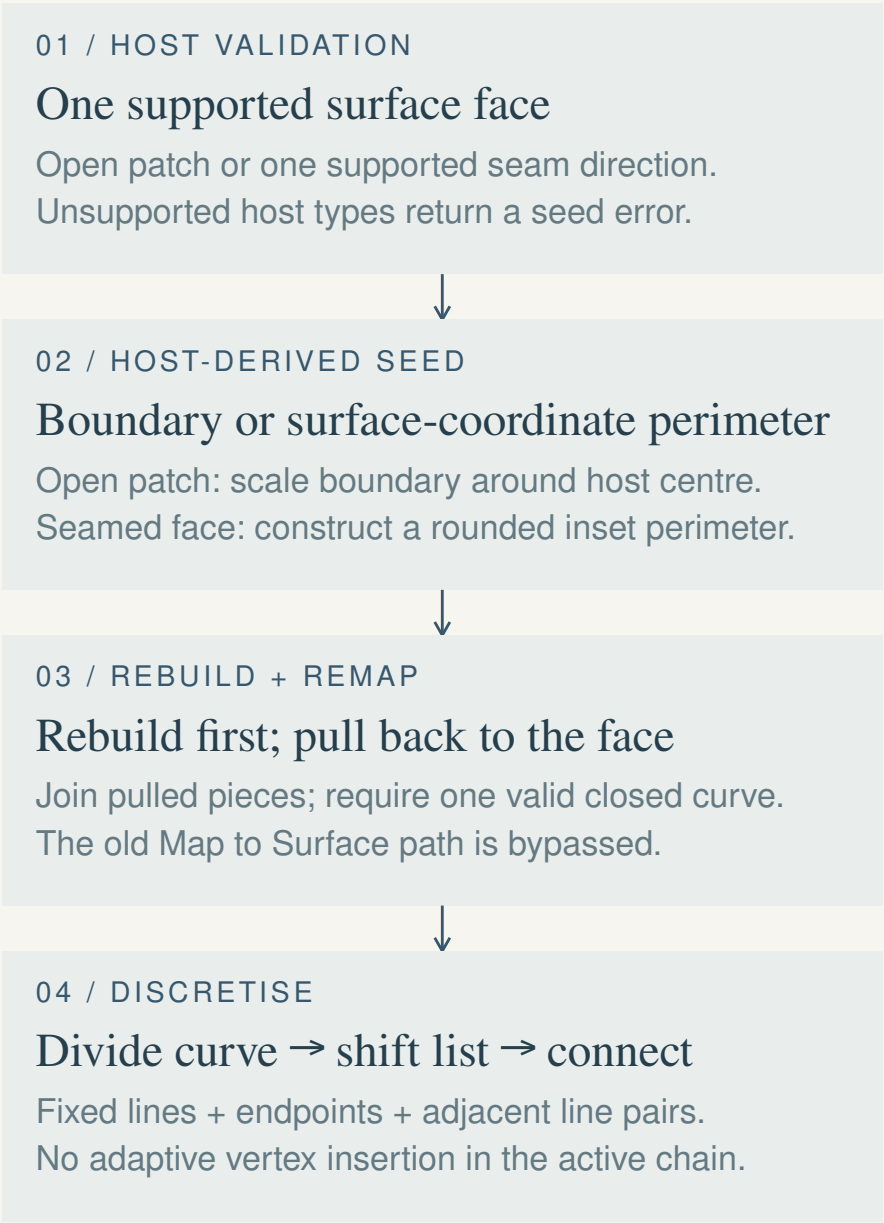
The definition verifies connections, scripts and saved controls.  
Checks estimate curvature and mesh deviation from 1,000 intervals.  
Coverage and overlap tests are not implemented.

HOW TO READ Follow 01–04 left to right; read each phase downward. Arrows carry data. The only feedback loop is inside the solver.

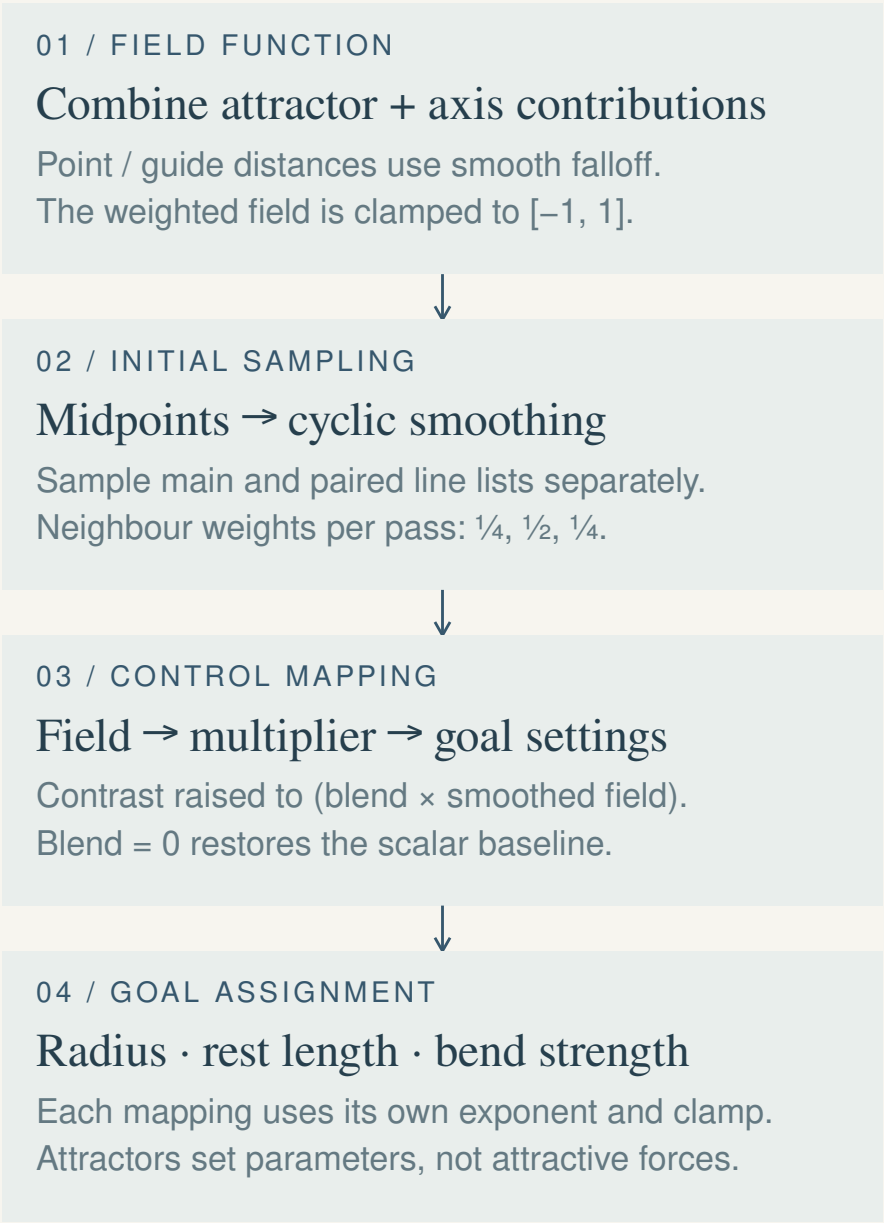
# Inside the workflow

What is prepared once, what is repeatedly solved, and what is checked only after pausing.

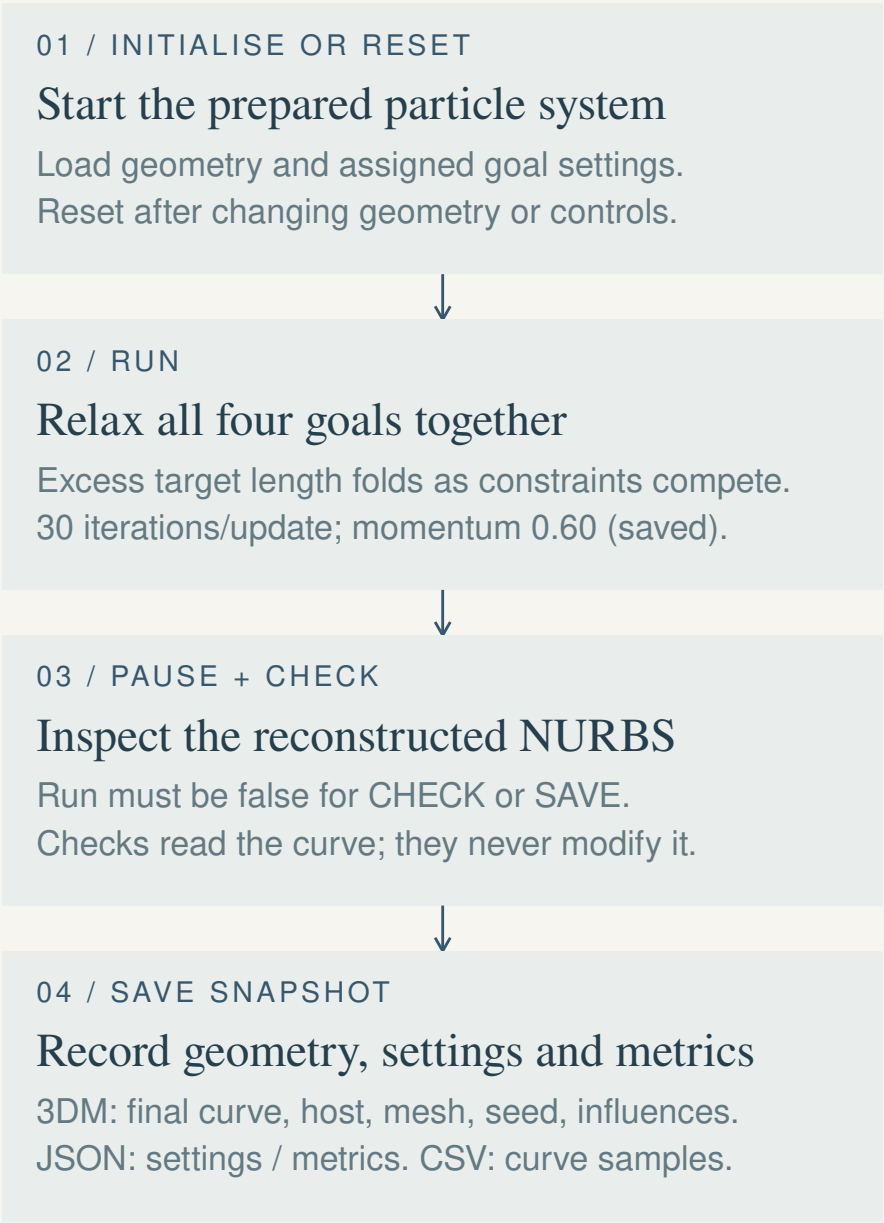
A / PREPARE THE NETWORK



B / TRANSLATE THE FIELD



C / RUN, CHECK + RECORD



## Why preparation matters

Pulling after rebuilding restores surface placement.  
Constraints act on a discretised network; the final smooth curve needs its own downstream checks.

## Why the preview can differ

Preview samples mesh vertices before smoothing.  
Solver assignments move with the original segments.  
A world-space field is not continuously re-applied.

## What a saved run can support

Closure and length; sampled bend radius / deviation.  
No automated coverage or overlap certification.  
Saved sliders are not verified for every supplied export.